

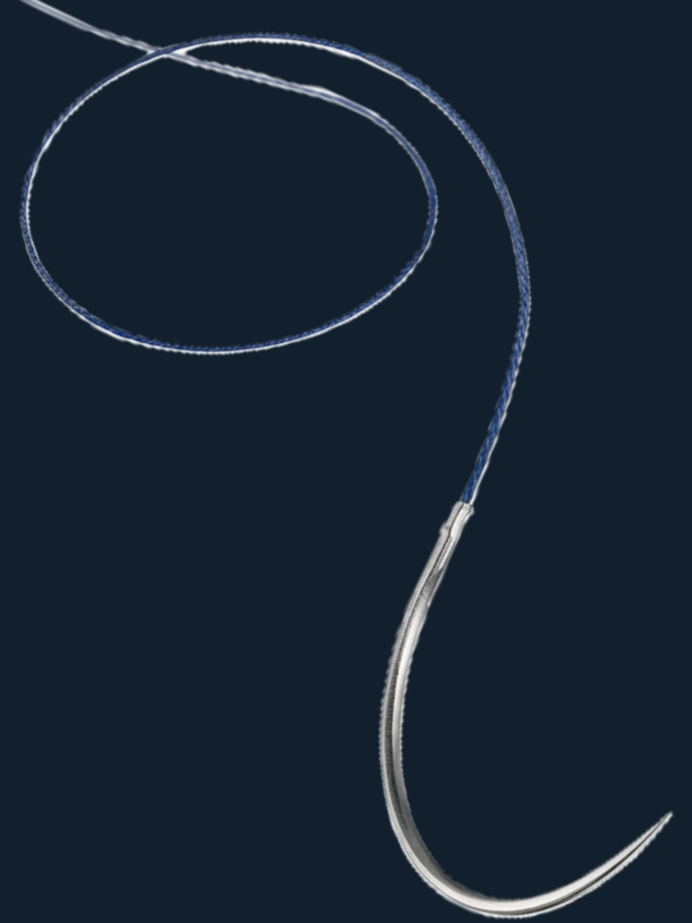


WundCheck

A mini CDSS that puts the daily wound check in the hands of the patient

Feline Weger, B.Sc. Biomedical Engineering

Sivanajani Sivakumar, B.Sc. Medical Informatics



HEALTH SYSTEM NEED

Surgical site infections appear after discharge

when nobody is looking

60.1 %

of surgical site infections occur after discharge — systematic review of 1.4 million operations. The CDC surveillance window runs 30 days, 90 for implants.

Too late

Warning signs are read as normal. The infection progresses; treatment starts days later than it could.

Too early

Normal healing signs — a scab, day-2 redness — trigger calls that bind practice staff and worry patients.

SSI incidence in Europe 0.5–10.1 % depending on procedure (ECDC 2017: 10,149 SSIs in 648,512 operations) · 9.5 % after colon surgery, 1.0 % after hip replacement (WHO) · Switzerland: > 170 hospitals in the Swissnoso surveillance programme

Why a knowledge-based CDSS — and not the alternatives

Information leaflet	already exists everywhere; no feedback on the actual finding
Telephone follow-up	not deliverable daily; “the wound looks odd” is not usable data
Photo-based ML	light, angle and skin tone vary; decisions not explainable
Rule-based CDSS	traceable to a guideline, reviewable, maintainable by clinicians

Scope

Adults · elective surgery · wounds with primary closure · any body region.

Not covered

Chronic wounds · burns · secondary healing · children · contaminated emergencies

Woelber et al., Surg Infect 2016;17(5):510–19 · ECDC Annual Epidemiological Report 2017 · WHO Global Guidelines for the Prevention of SSI · CDC/NHSN SSI Event

PERSONAS & SCENARIOS

Three users on three levels

only one of them operates the app



Peter Brunner, 58

Patient · primary user

Knee replacement 4 days ago. Little medical knowledge, reading glasses, uses everyday apps.

“I want to know it is fine — without calling anyone.”



Sandra Meier, 34

Practice assistant · secondary

Triages calls and messages all day. High workload, no time for guessing games.

“I need to see in 10 seconds whether this needs an appointment today.”



Dr Anna Keller, 45

Surgeon · tertiary

Operates and runs follow-ups. Wants the trajectory, not a retrospective recollection.

“I want the full trend without opening every single answer.”

SCENARIO A

Day 6 — the scab

36.9 °C, pain 2/10 and better than yesterday — but a brown scab has formed, and Peter's brother once had a wound infection. WundCheck returns GREEN with “a scab is a normal sign of healing — do not pick it off”. Peter does not call the clinic. The entry still goes into the record, so Dr Keller sees an unbroken trajectory at the suture removal appointment.

The call that is avoided is worth as much as the one that is triggered.

Four sources, three roles *and one we deliberately discarded*

CDC / NHSN

Normative — definition

Superficial vs. deep incisional SSI, 30/90-day window, the five clinical signs

Surveillance definition

NICE NG125

Normative — management

Post-operative wound care; §1.1.3 requires patients be taught to recognise an SSI

Evidence-based guideline

Southampton score

Instrument

Visual grading 0–V — the anchor for our image selection

Validated instrument

Campwala et al. 2019

Evidence

Compares CDC, ASEPSIS and Southampton

Retrospective, n = 22

How good is our evidence?

Campwala et al.: n = 22, retrospective, single-centre study in implant-based breast reconstruction. ASEPSIS showed better predictive performance than Southampton for implant failure.

→ **Southampton as a standardized assessment and communication framework rather than a predictive model.**

Why we discarded ASEPSIS — the best performer

ASEPSIS needs more data:
antibiotic administration, drainage, debridement, inpatient days and culture results.

→ **Not obtainable by a patient at home.**

Our image selection based on a published grading — *not arbitrary UI design*

What this buys us

- a citable source for our core interaction
- a non-trivial derived element
- an honest, evidenced limitation

The adaptation problem

Southampton grades “> 2 cm along the wound”. Patients do not measure. We ask “one small spot or the whole wound?” — closer to usable, further from precise.

0

Normal healing

→ Calm & dry

I

Mild bruising and/or erythema

→ Slightly reddened

II

Erythema plus other signs of inflammation

→ Markedly reddened & swollen

III

Clear / serous / bloody discharge

→ Clear or bloody discharge

IV

Pus

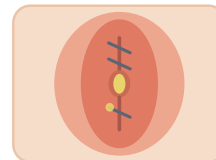
→ Cloudy / purulent discharge

V

Deep infection ± tissue breakdown

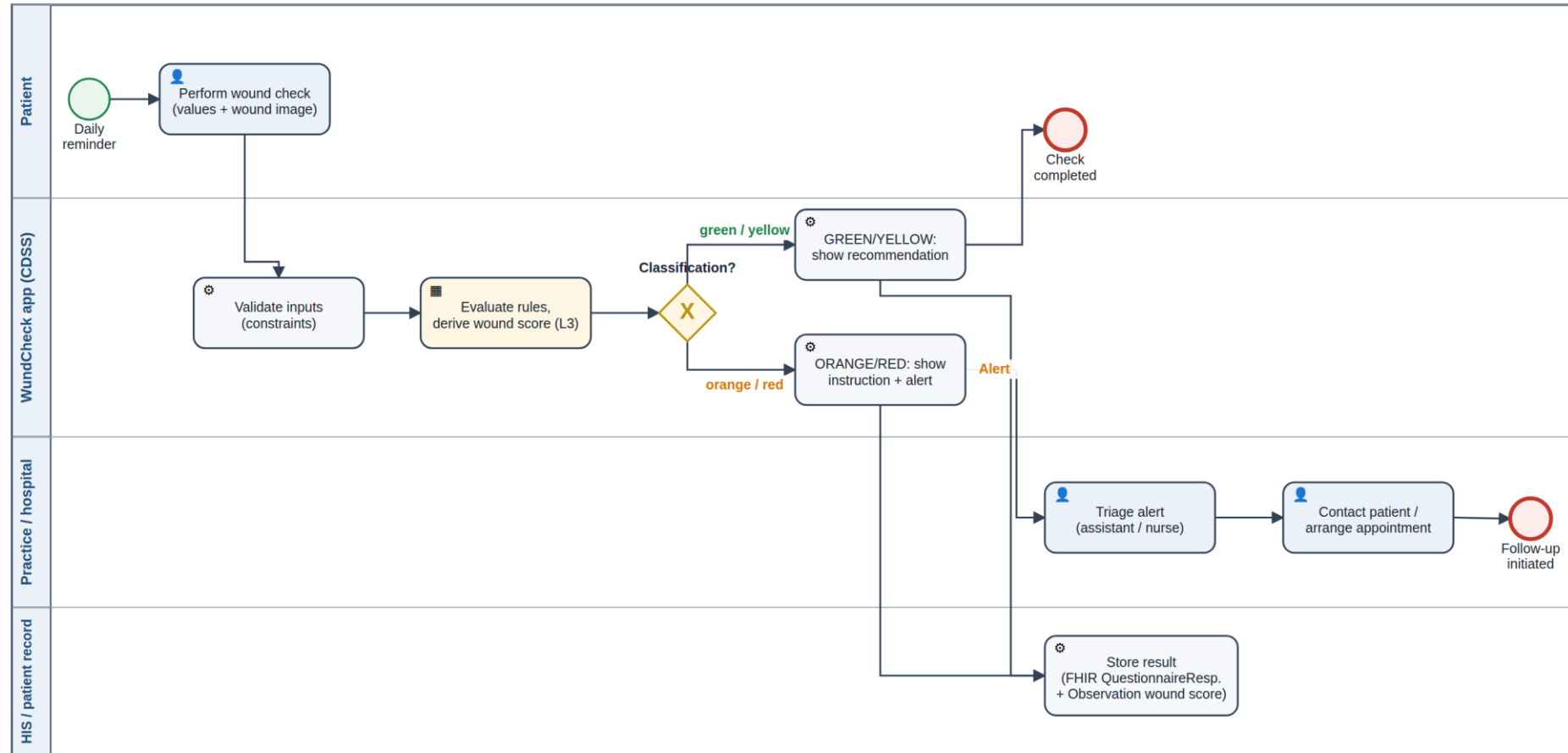
→ Open / pus

Examples:



Clinical decision logic is separated from the user interface

WundCheck — High-level workflow (BPMN): daily wound check after surgery with primary closure



Legend: 👤 User Task · ⚙️ Service Task (automated) · 📊 Business Rule Task (decision table L3) · X = exclusive gateway — editable source: workflow-highlevel.bpmn (bpmn.io / Camunda Modeler)

49 data elements —

everything traceable to a guideline

49

data elements

10

derived

38

terminology-bound

48

FHIR-mapped

ID	ELEMENT	TYPE	TERMINOLOGY	FHIR	L1 SOURCE
DE-020	Body temperature	decimal	LOINC 8310-5	Observation.valueQuantity	CDC — fever > 38 °C
DE-030	Wound appearance	select_one	SNOMED 225552003	Observation.valueCodeable	Southampton 0–V
DE-032	Red streak	select_one	SNOMED 46117009	Observation.valueCodeable	NICE NG125 §1.4.9
DE-035	Discharge	select_one	SNOMED 449736006	Observation.valueCodeable	CDC — purulent drainage
DE-045	Southampton grade	calculated	SNOMED 225552003	Observation.valueCodeable	Southampton

“Worst-first”: a reassuring finding can never overwrite a warning

RED · emergency

R1 R2 R3

Systemic signs · lymphangitis · dehiscence or pus



ORANGE · suspected

R9 R4 R5 R11 R12

Feeling unwell · purulent discharge · fever · persistence



YELLOW · abnormal

R6b R6a R7 R10

Inflammation · wound gap · bleeding on anticoagulants



GREEN · all clear

R8

Catch-all — never “everything is fine” without a contact hint

Why worst-first

A false alarm costs a phone call. False reassurance can delay an infection by days. We accept a higher false-alarm rate on purpose.

Two-pass evaluation (Same day)

Some rules depend on previous results (e.g. two consecutive yellow days). Therefore, the decision table is evaluated twice. The second pass can only escalate the urgency level.

All thresholds are named constants in the L3 with their source — a guideline change is one line, not a release.

Nine places where the guideline left us on our own

C-01

Professional instrument → layperson instrument

C-02

Sub-grades without centimetres

C-03

When does redness stop being normal?

C-04

Scab: warning or reassurance?

C-05

CDC criteria assume a clinical examination

C-06

Southampton as triage, not prognosis

C-07

Two fever thresholds instead of one

C-08

No trajectory on the first day

C-09

Severity scale ≠ urgency of action

The two highest-risk decisions sit in the same place

C-01 Southampton was built for trained staff. We turned it into an image selection for patients without medical knowledge. That is our core idea — and it is unvalidated.

C-03 No source defines the day on which redness stops being physiological. We set day 3. Too early means false alarms; too late means delayed detection.

All nine would need clinical sign-off. We do not have it — and we list that as a limitation.

Author in readable JSON, speak standards at the interface

XLSForm / ODK

Established, executable by existing engines

Standard widgets cannot carry our image cards — the L4 would dictate the UX

FHIR Questionnaire + CQL

Full standard, WHO SMART compliant

“FHIR resources are not human readable” — not reviewable by our clinical lead in six days

Hardcoding in the app

Fastest to a prototype

Every rule change needs a developer and a rebuild

Custom JSON + FHIR at the edge

Reads like the L2; full control of the UX; standards where they count

No existing engine runs it directly; expression language is not CQL



And we can prove it: a check scans the app source for every element id, threshold, code system and endpoint. Any clinical content in the L4 fails the build.

DEMO

Short Demo



Scan for the Demo

Study prototype — not for clinical use.

QUALITY ASSURANCE

Two defects were invisible until the rules actually ran

● B-01	Marked inflammation on day 1–2 classified GREEN	L2
● B-02	“I feel really unwell” triggered nothing at all	L2
● B-03	Emergency rule unreachable behind conditional logic	L2/L3
● B-04	Fever threshold duplicated in two places	L3
● B-05	Anticoagulant status collected but never used	L2
● B-06	Southampton IV mapped RED in one doc, ORANGE in another	L2
● B-07	Trend rule could never fire — circular dependency	L3/L4
● B-08	That same rule was unreachable even after the fix	L2

19/19

decision-logic tests

20/20

FHIR conformance checks

6

hazards ranked

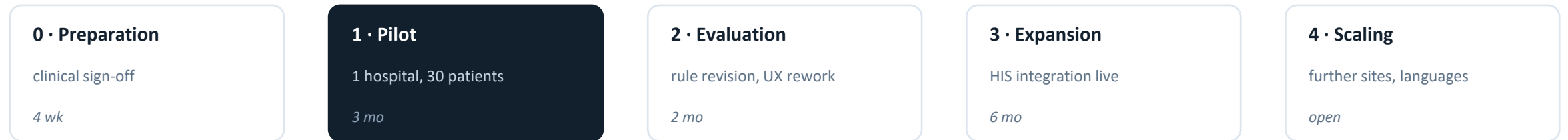
Residual risk

We never tested the one thing our whole approach rests on: whether laypeople assign the images correctly. That is monitored after deployment — GREEN cases that still present to a physician within 72 h.

Risk-based testing per Day 5: identify hazards → rank by risk → test critical paths first → document residual risk.

IMPLEMENTATION, MONITORING & EVALUATION

How it reaches users, how it is maintained & how we would know it works



Implementation

Provision QR code on the discharge sheet — no app store, no account, no install
Training Patient 5 min at discharge (first check done together) · practice 30 min · onboarding module for new staff
Maintenance Clinical department owns the content; four-eyes sign-off; guideline update to live in ≤ 4 weeks
Integration FHIR R4 to the HIS / Swiss EPD; endpoint is L3 configuration

Monitoring

Functionality > 99 % error-free runs
Stability zero data loss on network drop
Fidelity > 70 % of days checked
Data quality < 5 % incomplete entries

Evaluation

Pilot SUS + patient-vs-clinician agreement (κ)
Demonstration chart review — time to medical contact
Scale-up stepped-wedge; SSI rate, readmissions
Adverse effects false alarms, alert overrides, equity

Interoperability reached: structural yes · semantic partly · organisational no. · Most important safety indicator: GREEN cases that still present to a physician within 72 h.

False reassurance weighs more than a false alarm

Benefits

- Earlier treatment when an infection develops
- Fewer contacts without clinical consequence
- Complete trajectory for the follow-up appointment
- Structured, standardised follow-up data

Risks

- Automation bias — the patient stops observing
- Alert fatigue in the practice
- Scaling our own errors consistently, at volume
- Digital divide — the gap may widen

Is WundCheck a medical device? Yes — SaMD, IMDRF Category II

State of the healthcare situation = serious: an untreated SSI is not self-limiting and can lead to sepsis or revision surgery. Significance of information = drive clinical management: we neither diagnose nor treat, but we determine whether and when clinical action is taken.

Standards that would apply: ISO 14971 · IEC 62304 · IEC 62366 · ISO 13485. The EU AI Act does not — the system is rule-based, a deliberate consequence of avoiding image analysis.

	Inform	Drive	Treat/dx
Critical	IV	III	II
Serious	III	II	I
Non-serious	II	I	I

IMDRF risk categorisation

Individual contributions



Feline Weger

BSc Biomedical Engineer · clinical lead & L2

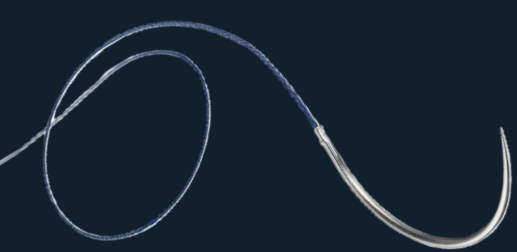
- L1 source research and evidence appraisal — CDC, NICE NG125, Southampton, Campwala et al.
- Health system need, scope decision, three personas and the scenarios
- Southampton → image-card mapping; the nine documented L1 → L2 translation decisions
- BPMN high-level and detail process · data dictionary (49 elements) · decision table (18 rules)



Sivanajani Sivakumar

BSc Medical Informatics · technical lead & QA

- L3 schema and authoring — thresholds, terminology bindings, FHIR mappings, versioning
- L4 application, expression engine, bilingual rendering, FHIR R4 transaction interface
- Hazard analysis, 19 decision-logic tests, 20 FHIR checks, the no-hardcoding check, bug log
- Implementation and M&E strategy · expected impact · SaMD classification



Everything is in the repository

Documents, BPMN sources, the L3, the running L4, the test and conformance scripts — all of it, reproducible.

github.com/Sivanajani/smart-wound-guideline-app

docs/ 01–14

every assignment item as its own document

l3/wundcheck-l3.json

the machine-readable CDSS — 49 elements, 18 rules

l4/wundcheck-app.html

one file, opens offline in any browser

tools/

19 logic tests · 20 FHIR checks · the no-hardcoding check

Thank you — questions?



Scan for the full project

Study prototype — not for clinical use.



Scan for the Demo

Study prototype — not for clinical use.

Sources

Woelber E et al.	Proportion of Surgical Site Infections Occurring after Hospital Discharge: A Systematic Review. Surg Infect 2016;17(5):510–19.	60.1 % post-discharge
ECDC	Healthcare-associated infections: surgical site infections — Annual Epidemiological Report 2017.	0.5–10.1 % by procedure
CDC / NHSN	Surgical Site Infection Event (SSI). Patient Safety Component Manual.	30 / 90-day window
NICE	Surgical site infections: prevention and treatment. NG125, 11 Apr 2019, upd. 19 Aug 2020.	nice.org.uk/guidance/ng125
Bailey IS et al.	Southampton Wound Scoring System.	grades 0–V
Campwala I, Unsell K, Gupta S	A Comparative Analysis of Surgical Wound Infection Methods: CDC, ASEPSIS and Southampton in breast reconstruction SSI. Plastic Surgery, 2019.	PMID 31106164
WHO	Monitoring and Evaluating Digital Health Interventions: a practical guide. 2016.	maturity stages
WHO	SMART Guidelines — the L1–L5 knowledge layers.	who.int
IMDRF	Software as a Medical Device: possible framework for risk categorisation.	categories I–IV
HL7	FHIR R4 — QuestionnaireResponse, Observation, Communication.	hl7.org/fhir/R4

Decision table — all 18 rules

R1	temp ≥ 39 OR chills = 1	CDC · sepsis red flags
R2	red_streak = 1 OR redness_extent = 4	NICE NG125 §1.4.9
R3	wound_open = 2 OR bleeding = 2 OR wound_state = 4	CDC deep incisional
R9	wellbeing = 2 OR nausea_vomiting = 2	systemic signs
R4	secretion = 3 OR secretion_smell = 1	CDC superficial incisional
R5	fever OR (pain ≥ 7 AND worsening AND day ≥ 3)	CDC · NICE
R11	secretion ≥ 1 AND discharge ≥ 3 days	Southampton IIId
R12	classification = yellow AND yesterday = yellow	persistence
R6b	Southampton II OR swelling = 3 (from day 1)	Southampton II
R6a	Southampton I AND day ≥ 3	wound healing phases
R7	wound_open = 1 OR numbness OR mobility change	follow-up standards
R10	bleeding ≥ 1 AND anticoagulants	bleeding assessment
R8	catch-all	—
I1–I5	risk factors · medication · smoking · scab · high-risk procedure	non-exclusive

Test protocol — 19 decision-logic cases, all passing

T1	R1 fever · temp 39.2	RED (R1)	✓	T11	B-02 fixed · feeling unwell	ORANGE (R9)	✓
T2	R1 boundary · temp 39.0	RED — inclusive	✓	T12	B-03 fixed · red streak	RED (R2)	✓
T3	R3 safety · image “open/pus” alone	RED (R3)	✓	T13	B-01 fixed · grade II on day 2	YELLOW (R6b)	✓
T4	R5 before R6	ORANGE (R5)	✓	T14	B-01 counter-check	GREEN	✓
T5	mild signs on day 2	GREEN	✓	T15	Southampton derivation	grade IV → ORANGE	✓
T6	conditional logic	3 follow-ups hidden	✓	T16	B-05 fixed · bleeding + anticoag	YELLOW (R10)	✓
T7	constraint · temp 45	input rejected	✓	T17	persistence rule	ORANGE (R12)	✓
T8	info rules non-exclusive	GREEN + I1 + I2 + I3	✓	T18	discharge ≥ 3 days	ORANGE (R11)	✓
T9	scab reassures	GREEN + I4	✓	T19	threshold change without rebuild	GREEN → ORANGE	✓
T10	worst-first vs reassurance	ORANGE (R4) + I4	✓				

Reproduce: `node tools/run-tests.mjs` · `node tools/check-fhir.mjs` · `node tools/check-hardcoding.mjs`

Hazard analysis and risk matrix

H-01	An infection is classified GREEN	sev 5 · lik 3	15
H-02	An emergency is classified ORANGE	sev 5 · lik 2	10
H-03	The patient abandons the check	sev 3 · lik 3	9
H-04	False alarm during normal healing	sev 2 · lik 4	8
H-06	The practice overlooks an alert	sev 4 · lik 2	8
H-05	An implausible input is processed	sev 3 · lik 2	6

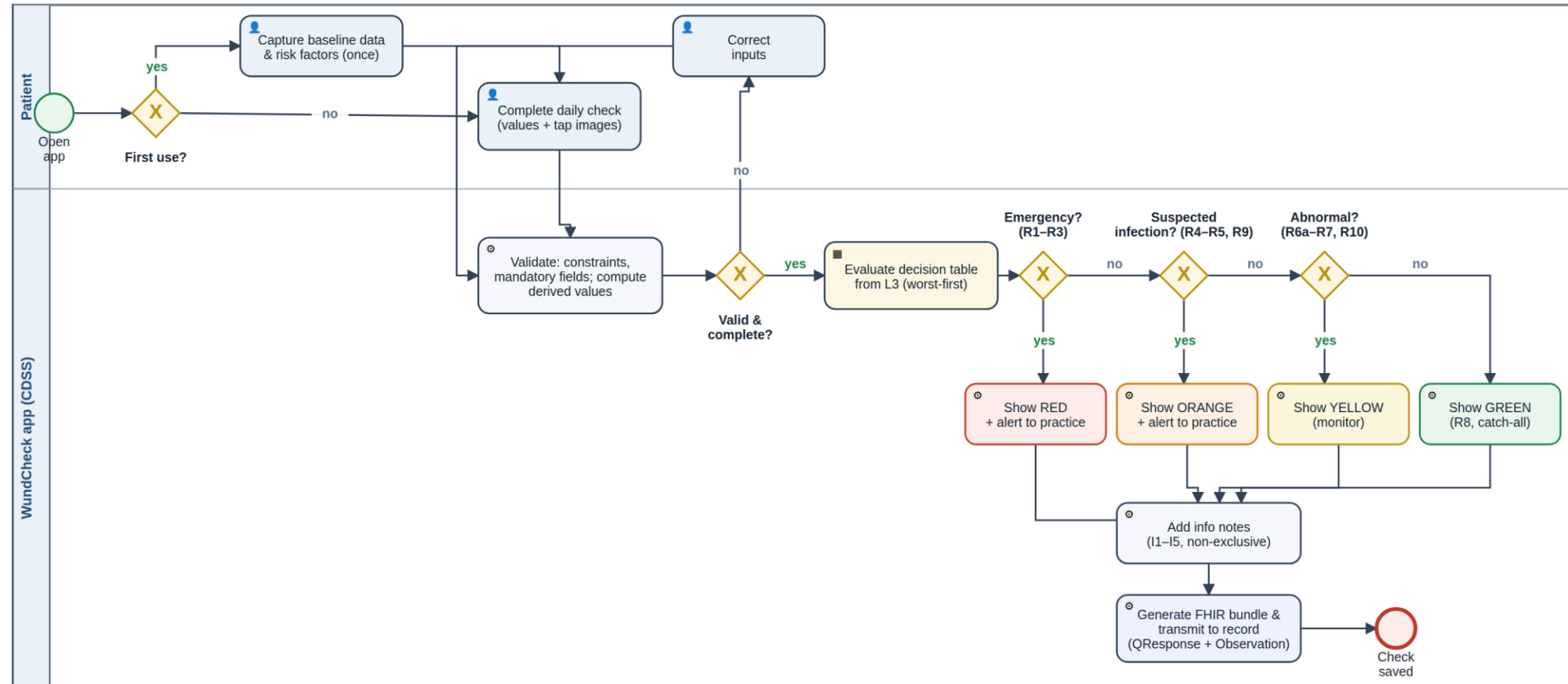
Severity

5	5	H-02	H-01	20	25
4	4	H-06	12	16	20
3	3	H-05	H-03	12	15
2	2	4	6	H-04	10
1	1	2	3	4	5
	1	2	3	4	5
	Likelihood				

Score ≥ 10 → full test coverage. Score 6–9 → sample-based coverage.

Detailed process inside the app

WundCheck app — detailed process of one daily check (BPMN): validation, worst-first decision logic, FHIR transmission



Gateway cascade = decision logic in worst-first order (safety principle): emergency before suspected infection before abnormal; GREEN only as catch-all. Editable source: workflow-app-detail.bpmn

The FHIR interface — a real transaction bundle, not a download

QuestionnaireResponse

every answer, typed and terminology-coded; canonical carries the L3 version

every run

Observation

traffic light, Southampton grade, triggering rule, L3 version, days post-op

every run

Communication

structured alert, priority urgent (ORANGE) or stat (RED)

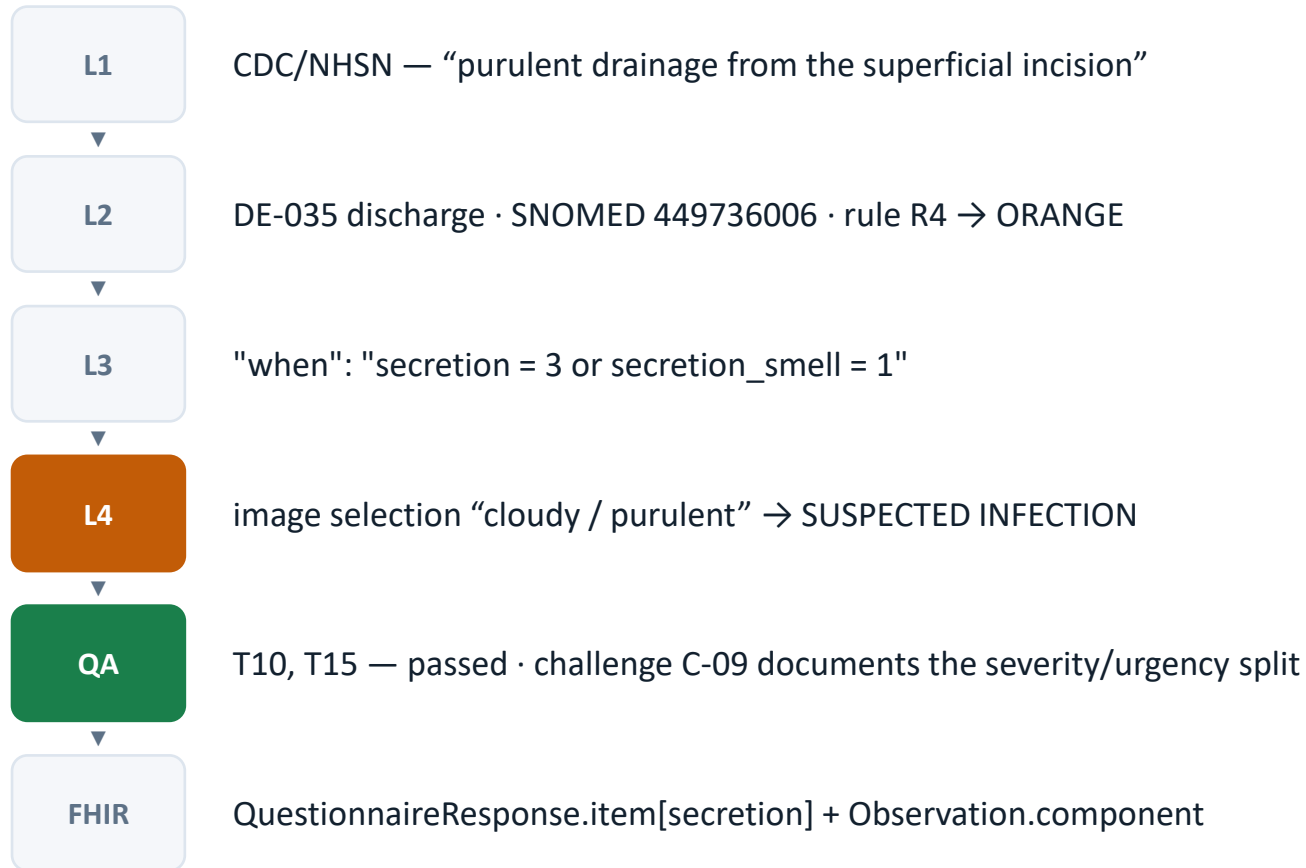
ORANGE / RED only

The endpoint is configuration, not code

```
"fhir": { "endpoint": "https://.../baseR4", "patient_reference": "Patient/example",
          "alert_recipient": "Organization/practice", "send_mode": "transaction" }
```

Redirecting to a hospital FHIR server or the Swiss EPD gateway is one line of JSON. If the endpoint is unreachable the bundle queues — the patient still gets the recommendation.

Traceability — one finding, all the way through



Every element, rule, challenge, hazard, test and defect carries an ID that is identical in all artefacts.

The L3 in one screen — this file is the CDSS

```
{ "meta": {
  "version": "0.3.3",  "languages": ["en","de"],  "default_language": "en",
  "l1_sources": [ CDC/NHSN · NICE NG125 · Southampton · Campwala 2019 ],
  "thresholds": { "fever_orange": 38.0,    // CDC – deep incisional SSI
                  "fever_red":    39.0,    // group decision – challenge C-07
                  "normal_healing_days": 3 }, // group decision – C-03
  "fhir": { "endpoint": ".../baseR4", "send_mode": "transaction" } },

"elements": [ {
  "id": "secretion",  "de_id": "DE-035",  "type": "select_one",
  "label": { "en": "Is fluid coming out of the wound?",
             "de": "Kommt Flüssigkeit aus der Wunde?" },
  "options": [ 0 = none · 1 = clear · 2 = bloody · 3 = purulent, each with an image ],
  "terminology": { "snomed": "449736006" },
  "fhir": "Observation.valueCodeableConcept",
  "l1_source": "CDC/NHSN – purulent drainage · Southampton III-IV" } ],

"decision_rules": [ {
  "id": "R4",  "priority": "orange",  "exclusive": true,
  "when": "secretion = 3 or secretion_smell = 1",
  "output": { "en": "SUSPECTED INFECTION – contact your practice today..." },
  "annotation": "Purulent drainage is a primary criterion of superficial SSI.",
  "reference": "CDC/NHSN superficial incisional SSI · Southampton IV" } ] }
```

Bilingual by construction

Every user-facing string is a language-keyed object. A further language is a content change, not a code change.

Thresholds carry their source

Each constant names where it comes from — a guideline value or a documented group decision.

Rules carry their reference

Every rule states its L1 source and an annotation, so a clinician can review it without reading code.

The endpoint is configuration

Redirecting to a hospital FHIR server is one line — no code change, no rebuild.

49 elements · 18 rules · 74 KB · read by the L4 at runtime — swap this file and the app becomes a different CDSS.

Data dictionary — the four required columns

ID	LABEL	TYPE	RESPONSE OPTIONS	CONDITIONAL LOGIC	CONSTRAINT	TERMINOLOGY
DE-020	Body temperature today	decimal	free numeric input, °C	—	34–43 °C	LOINC 8310-5
DE-023	Pain 0–10	integer	NRS 0–10	—	0–10	LOINC 72514-3
DE-026	How do you feel overall?	select_one	0 = well · 1 = tired · 2 = really unwell	—	—	SNOMED 386661006
DE-030	Which picture matches your wound?	select_one	1 = calm & dry · 2 = slightly red · 3 = markedly red & swollen · 4 = open / pus	—	—	SNOMED 225552003
DE-032	Does a red streak run away?	select_one	1 = yes · 0 = no	none — deliberately ungated (B-03)	—	SNOMED 46117009
DE-035	Is fluid coming out?	select_one	0 = none · 1 = clear · 2 = bloody · 3 = cloudy / purulent	—	—	SNOMED 449736006
DE-036	Does the discharge smell?	select_one	1 = yes · 0 = no	secretion ≥ 1	—	SNOMED 288227007
DE-037	One spot or the whole wound?	select_one	1 = small spot · 2 = along the wound	secretion ≥ 1	—	—
DE-038	For how many days?	integer	free numeric input, days	secretion ≥ 1	0–60	—
DE-025	Did pain disturb your sleep?	select_one	1 = yes · 0 = no	pain ≥ 4	—	SNOMED 301345002
DE-045	Southampton grade	calculated	0 · I · II · III · IV · V	derived from 5 elements	—	SNOMED 225552003

49 elements in total · 10 derived · 4 with conditional logic · 8 with constraints · 38 with a terminology binding · 48 with a FHIR mapping. Full table: docs/05.

Depth that did not fit into ten minutes

Testing types applied (Day 5 vocabulary)

- L2 verification — decision table $\leftrightarrow$ BPMN cascade $\leftrightarrow$ L3 rules reconciled
- Functional testing — individual rules, exclusions, severity order, boundary values
- Integration testing — derived elements $\rightarrow$ rules $\rightarrow$ output; conditional logic $\rightarrow$ rule reachability
- Usability testing — image assignment by lay people: NOT carried out, our largest residual risk

The L3 implementation package (Day 3)

- Documentation — health need, personas, BPMN
- Content / workflow — sections, relevant, decision rules
- Terminology — SNOMED CT / LOINC / ICD-10 bindings
- Validation — 19 test cases, test data, conformance checks
- Interfaces — FHIR export · UI/UX — image options, traffic light

Stakeholders and where it could fail

- The critical group is the practice, not the patient
- If Sandra experiences the alerts as a burden, the rollout fails — however good the classification is
- Hence: alerts only from ORANGE upwards, structured instead of free text, alert load measured explicitly

What we would do with three more months

- 1 · Clinical sign-off on every threshold and rule — it settles all nine L1 $\rightarrow$ L2 decisions at once
- 2 · Usability study on the image assignment — validates the central design idea
- 3 · Graphics for several skin tones and body regions
- 4 · Convert the L3 to a FHIR Questionnaire — the software neutrality we consciously gave up